Attention-Deficit/Hyperactivity Disorder and Electroencephalography: A Concise Review

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***Abstract*—Attention-Deficit/Hyperactivity Disorder (ADHD) affects about 5–7% of children worldwide and is marked by ongoing problems with attention, hyperactivity, and impulsive behavior that may continue into adulthood. Although diagnosis is mainly based on behavior, research increasingly links ADHD to differences in brain development and neural activity. Electroencephalography (EEG) is widely used in ADHD research because it is non-invasive, suitable for children, and captures brain activity with high temporal resolution. These include increased theta activity, decreased beta activity, modified theta-beta ratios, and alterations in attention and error-related responses, such as P300, N2, error-related negativity (ERN), and contingent negative variation (CNV). Disrupted communication within attention-related brain networks has also been observed. Machine learning methods, including support vector machines, ensemble approaches, and deep learning architectures such as convolutional neural networks (CNNs) and transformers, have been applied to EEG-based ADHD classification. However, clinical translation remains limited by small datasets, inconsistent preprocessing pipelines, and limited model interpretability.**

***Keywords*—ADHD, EEG, spectral power, ERPs, connectivity, biomarkers, machine learning**

I. Introduction

ADHD is one of the most common neurodevelopmental disorders, affecting an estimated 5–7% of children worldwide as well as a significant number of adults [1]. It is characterized by difficulties with attention, hyperactivity, and impulsive behavior, which can have long-term consequences for cognitive development, academic performance, emotional regulation, and psychosocial functioning across the lifespan [1]. While diagnosis is still largely based on behavioral criteria, growing evidence suggests that ADHD is linked to atypical brain development and altered organization of large-scale neural networks [2]. As a result, neuroimaging research has increasingly focused on identifying biological markers that could complement behavioral assessments and improve clinical understanding [3]. Electroencephalography (EEG) has emerged as a particularly valuable tool in ADHD research because it is noninvasive and well suited for capturing fast neural dynamics related to cortical arousal and cognitive processes such as attention, inhibition, and executive control [4]. A substantial body of research has reported EEG alterations in individuals with ADHD, including changes in spectral power, disrupted functional connectivity, and atypical event-related potential (ERP) components [5]. Early interpretations of these findings often emphasized cortical underarousal or delayed brain maturation, typically reflected in increased slow-wave activity. More recent work, however, highlights disruptions in frontoparietal and cognitive control networks as key neural mechanisms underlying ADHD symptoms [5], [6]. Researchers use supervised and deep learning on EEG features (frequency power, connectivity, ERPs, or raw signals) to distinguish ADHD from typical individuals with improved accuracy. By identifying important EEG features, interpretable machine learning enhances model reliability and comprehension of ADHD-related brain activity [6]. However, due to differences in participant characteristics and study design, EEG results are still variable, which limits their clinical usefulness [7]. This review addresses current issues and potential future paths while summarizing important EEG findings.

II. Literature

ADHD comes with difficulties in attention, impulse control, and activity regulation which affects learning and daily functioning in individuals [8], [9]. Clinically, it is categorized into three presentations: combined, hyperactive–impulsive, and predominantly inattentive, reflecting differences in behaviour and underlying cognitive processes [10]. The causes of ADHD are complex and involve an interplay of genetic, neurobiological, and environmental factors. Evidence from twin and family studies consistently highlights a strong genetic contribution, with heritability estimates ranging from 70% to 80% [11]. Neurobiological research has repeatedly implicated frontostriatal and frontoparietal networks that support executive control, attention regulation, reward processing, and inhibitory control [12]. Findings of delayed cortical maturation, particularly in prefrontal regions, further support the view of ADHD as a disorder of altered brain development rather than focal dysfunction [13]. Sleep EEG studies have similarly reported atypical slow-wave activity in children with ADHD, further reflecting disrupted neural maturation [14]. Environmental influences—including prenatal exposure to toxins, maternal stress, low birth weight, early adversity, and ongoing psychosocial stress—may further shape both risk and symptom severity [15]. More recent theoretical frameworks emphasize gene–environment interactions, framing ADHD as the result of atypical neurodevelopmental trajectories rather than abnormalities confined to a single brain region [11], [15]. At the cognitive level, ADHD is commonly associated with impairments in sustained attention, working memory, inhibitory control, and time perception. These difficulties are often accompanied by increased variability in reaction times and challenges in managing competing cognitive demands [16]. Learning disabilities, oppositional defiant disorder (ODD), anxiety, depression, and autism spectrum disorder (ASD) are common comorbid disorders that can affect symptom manifestation, treatment effects, and electrophysiological findings [17]. This broad clinical and cognitive heterogeneity is also reflected in EEG research, highlighting the importance of considering ADHD subtype, developmental stage, and comorbidities when interpreting neurophysiological results.

III. EEG and ADHD

Electrical activity produced mainly by the synchronous postsynaptic potentials of cortical pyramidal neurons is captured by electroencephalography (EEG), a noninvasive neuroimaging method [18]. EEG's millisecond-level temporal precision makes it ideal for researching the quick brain processes that underlie executive control, attention, and sensory processing—domains that are often compromised in people with ADHD.

EEG signals are typically analyzed across several domains:

A. Morphological Differences in EEG Signals Between ADHD and Neurotypical Subjects

ADHD EEG frequently shows increased theta and decreased beta activity, indicating less alertness, in addition to weaker P300, N2, and ERN responses, which indicate slower processing and impaired attention [4], [5]. Although these patterns are not diagnostic on their own, reduced frontoparietal connection further suggests poor network coordination and supports EEG-based ADHD identification.

B. Spectral Power Alterations

ADHD is often associated with increased theta and decreased beta activity, leading to a higher theta/beta ratio [4], [5], [19], however its diagnostic value varies. Although other band changes (gamma, delta, and alpha) are also linked to cognitive problems, their applicability as trustworthy clinical indicators is limited by the diversity among ADHD types [20]. EEG spectral patterns differ among ADHD categories, as Table I illustrates. Inattentive individuals exhibit more theta activity, while hyperactive-impulsive cases exhibit larger beta and gamma activity. Nevertheless, these markers are not specific enough to be used as a stand-alone diagnosis.

C. Event-Related Potentials (ERPs)

ERPs show lower and delayed P3 and N2, as well as weaker ERN, indicating deficiencies in attention, inhibition, and error monitoring in ADHD [3]–[5]. Further alterations in CNV and LPP indicate problems with task preparation and emotional processing, but consistency is limited by study variability (Please refer Table II).

D. Connectivity and Network Dynamics

EEG connectivity studies show that ADHD is associated with disruptions in large-scale brain networks, such as reduced frontoparietal connectivity and altered default mode network (DMN) activity, which affect attention and cognitive control [6] [7] [10]. Combining linear and nonlinear connectivity measures, including Pearson Correlation and Phase-Locking Value, can improve the detection of these abnormalities [21]. However, variations in datasets and analysis approaches lead to inconsistent findings (Table III).

E. Machine Learning and Deep Learning Approaches

Machine learning methods are increasingly used to identify EEG-based patterns linked to ADHD. Traditional models such as support vector machines, random forests, and logistic regression generally show moderate performance when using features from spectral power, ERPs, and connectivity, with better results often achieved by combining multiple feature types [32]. Among these, ERP-based features appear especially useful for capturing differences in cognitive processing and inhibitory control in ADHD . The performance of traditional machine learning models mainly depends on how well useful features are extracted from raw EEG signals. Commonly used features include: (1) spectral power, which measures brain wave activity in delta, theta, alpha, beta, and gamma bands, including the widely studied theta-to-beta ratio [4], [19]; (2) entropy and complexity features, such as approximate entropy and fractal dimensions, which capture irregular and nonlinear patterns in EEG signals ; (3) time–frequency features, including wavelet and spectrogram-based methods that analyze how signals change over time; (4) Hjorth parameters, simple statistical measures describing signal activity and variation; (5) connectivity features, which show how different brain regions interact through coherence or phase relationships; and (6) ERP features, such as amplitude and latency of P300, N2, and ERN components that reflect brain responses to stimuli. Feature selection techniques like LASSO or mutual information are often used to keep the most important features and reduce complexity. More recently, deep learning models, including convolutional neural networks (CNNs) and long short-term memory (LSTM) networks, have reported improved performance by learning spatial and temporal patterns directly from raw or minimally processed EEG data [24], [25].Notably, Chugh et al. [25] proposed a hybrid CNN-LSTM architecture evaluated on two public EEG datasets, achieving classification accuracies of 98.86% and 98.28%, respectively, outperforming several contemporary models. Transformer-based architectures have also shown strong promise, with models such as EEG-Transformer demonstrating competitive classification performance on EEG data from ADHD and control children [26]. However, their generalizability remains limited due to small sample sizes, varied preprocessing methods, site-specific

TABLE I
SPECTRAL POWER ALTERATIONS IN EEG REPORTED IN ADHD

Band	Change	Key Findings
Theta (4–8 Hz)	Increased	Underarousal / delayed maturation; inefficient attention allocation (often frontal-central). Basis of TBR.
Beta (12–30 Hz)	Decreased	Reduced task engagement and inhibitory control; contributes to elevated TBR.
TBR (Theta/Beta)	Increased (mainly children)	Age-dependent and affected by fatigue/comorbidity; inconsistent as a diagnostic marker.
Alpha (8–12 Hz)	Reduced suppression	Poor attentional modulation; frontal alpha asymmetry linked to emotional dysregulation.
Delta (0.5–4 Hz)	Increased (subgroups)	May reflect developmental delay or atypical cortical organization.
Gamma (>30 Hz)	Altered	Disrupted higher-order integration and working memory; less studied.
Subtype effects	Mixed	Inattentive: stronger theta; Hyperactive–impulsive: beta/gamma disruptions.

TABLE II
ERP ALTERATIONS REPORTED IN ADHD

Component	Change	Cognitive Implication
P3 / P300	Reduced amplitude; delayed latency; altered topography	Lower attentional allocation, weaker context updating, slower stimulus evaluation.
N2	Reduced amplitude; delayed latency	Impaired conflict monitoring and inhibitory control; increased impulsivity.
ERN	Attenuated amplitude; altered post-error adjustment	Weaker error monitoring (ACC-related), reduced learning from mistakes.
CNV	Reduced buildup; impaired expectancy/preparation	Deficits in anticipatory attention and motor readiness (proactive control).
LPP	Atypical modulation	Altered emotional processing and regulation (often with comorbid symptoms).

TABLE III
EEG CONNECTIVITY AND NETWORK ALTERATIONS IN ADHD

Domain	Abnormalities	Cognitive Implications
Frontoparietal	Reduced theta/alpha coherence; weaker long-range coupling	Impaired top-down attention control, sustained attention, and task switching.
Frontal (Executive)	Lower frontal theta sync; abnormal beta/gamma connectivity	Deficits in inhibitory control and motor regulation; increased impulsivity.
DMN Regulation	Poor task-related DMN suppression; elevated resting DMN connectivity	Mind-wandering and attentional lapses; weaker task engagement.
Temporal/Phase Dynamics	Reduced phase sync; weaker theta–gamma coupling; altered cross-frequency links	Disrupted oscillatory coordination affecting attention and working memory.
Graph Properties	Reduced small-worldness; altered modularity; lower global efficiency	Less efficient network organization and information transfer.

effects, and clinical heterogeneity [32]. In addition, limited interpretability continues to restrict clinical adoption, prompting current research to focus on more interpretable models, larger multisite datasets, multimodal integration with MRI and behavioral measures, and stronger cross-dataset validation to support reliable clinical translation. Apart from model design, several practical factors strongly influence the performance of EEG-based ADHD classification models. Proper hyperparameter tuning is important for improving accuracy, such as selecting suitable kernel and regularization values for SVM or adjusting the number of trees and learning rate in ensemble methods. In deep learning, factors like optimizer type, learning rate, batch size, dropout, and network depth also affect performance, but these details are not always clearly reported, making reproduction difficult [31]. Dataset-related issues such as class imbalance between ADHD and control groups can bias results, often requiring oversampling, undersampling, or class-weighted training. Small dataset size can also lead to overfitting, so regularization or transfer learning methods are often applied. Additionally, the evaluation method plays an important role, as subject-dependent testing usually gives higher accuracy than subject-independent evaluation. Therefore, future studies should clearly report evaluation strategies and include metrics such as sensitivity, specificity, and AUC along with accuracy for better comparison across studies.

1) Empirical Validation Across Recent Studies: Recent advances in machine learning and deep learning have substantially improved the accuracy of EEG-based ADHD classification. As shown in Table IV, research from 2016 to 2025 reflects a clear transition from traditional methods to more advanced architectures, including transformer models and deep learning models. Early studies using support vector machines and handcrafted spectral features typically reported accuracies of 78–85%, whereas more recent deep learning approaches commonly achieve accuracies above 88–98%. Classification accuracies of up to 98.86% have been reported on public benchmark datasets, while the best performance on clinical datasets reaches 95.59% [34]. However, these results are often based on relatively small datasets and varied EEG recording protocols, which may limit generalizability. Overall, the literature demonstrates a clear improvement in classification performance over the past years, driven by advances in feature representation, model architecture, and data augmentation strategies.

IV. Methodology

A. Search Strategy and Study Selection

A systematic search was conducted to collect studies on EEG-based ADHD analysis using machine learning. Databases such as IEEE Xplore, PubMed, Scopus, and Google Scholar were searched for papers published between 2016 and 2025 using keywords related to ADHD, EEG, machine learning, and deep learning. A total of 420 records were identified from database searches, with an additional 35 records from other sources. After duplicate removal, 350 records remained for screening, of which 90 full-text articles were assessed for eligibility.

Inclusion criteria: peer-reviewed studies involving human ADHD participants, use of EEG data, application of machine learning or deep learning methods, and reporting of performance metrics such as accuracy or AUC.

Exclusion criteria: studies without EEG data, review papers, simulated data studies, and non-English publications.

After removing duplicates, relevant papers were selected through title, abstract, and full-text screening. The final studies are summarized in Table IV, and the selection process is shown in the PRISMA diagram (Fig. 1). Table IV shows a carefully selected set of representative studies from the overall collection to highlight the performance trends across different model categories. It is not meant to include every study, but instead provides a clear and simplified overview of how various approaches perform in comparison.

V. Comparative Analysis of Machine Learning Approaches

To better understand how different machine learning techniques perform in EEG-based ADHD classification, we compared reported accuracy values across four major categories: Traditional Machine Learning, Boosting Algorithms, Deep Learning Models, and Advanced Architectures. The comparison is based on the studies summarized in Table IV, which includes research published between 2016 and 2025.

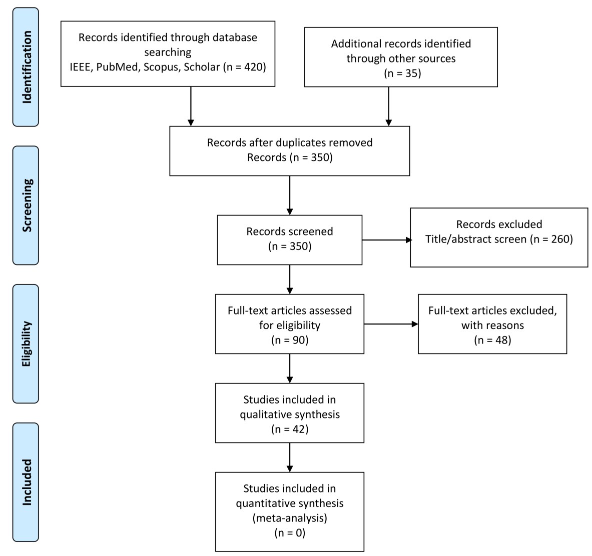

Fig. 1. PRISMA 2009 flow diagram of study selection for the ADHD EEG-based classification review.

A. Performance Across Machine Learning Categories

Machine learning approaches for EEG-based ADHD classification have evolved considerably over recent years, shifting from conventional statistical models to more advanced neural network architectures. To enable a fair comparison, the average classification accuracy for each category was computed using the studies listed in Table IV.

Calculation Methodology: The average performance for each category was obtained using the arithmetic mean of reported accuracy values, giving equal importance to each model. The Traditional Machine Learning category includes commonly applied methods such as Support Vector Machine (SVM) [36], Random Forest [27], and Logistic Regression [33], which rely on manually extracted EEG features like spectral power and statistical measures.

The Boosting Algorithms category consists of XGBoost [22] and LightGBM [23], which improve prediction performance by combining multiple weak learners. The Deep Learning category includes models such as CNN–LSTM architectures [37] and other neural network approaches reviewed in [32], which automatically learn spatial and temporal patterns from EEG signals.

The Advanced Architectures category covers recent techniques such as Transfer Learning-based CNN models [38] and Transformer-based hybrid frameworks [34], which are designed to capture complex relationships and long-range dependencies in EEG data more effectively.

B. Key Findings from Category Comparison

Table IV shows clear differences in performance among the machine learning approaches used for EEG-based ADHD classification. Advanced architectures achieved an average accuracy of 95.30% (range: 95.0–95.59%), with Alsharif et

TABLE IV
SELECTED REPRESENTATIVE EEG-BASED ADHD CLASSIFICATION STUDIES.

Category	Study	Method	Acc.%	Dataset	Features
Traditional ML	Guo et al. (2022) [36]	Autoencoder + SVM	89.10	IEEE DataPort [30]	Latent features
Boosting	Kim et al. (2025) [35]	XGBoost	90.80	Custom clinical	Spectral + nonlinear
Deep Learning	Wang et al. (2022) [37]	CNN–LSTM	93.70	Custom clinical	Spectrogram
	Chugh et al. (2024) [25]	CNN–LSTM	98.86	IEEE DataPort [30]	EEG spatial + temporal features
	Dubreuil-Vall et al. (2020) [24]	CNN(4-layer)	88.00	Custom clinical	EEG spectrogram
Advanced Architectures	Aldayel et al. (2023) [38]	Transfer Learning CNN	95.00	Custom clinical	Cross-age features
	Alsharif et al. (2024) [34]	CNN-BiLSTM-GRU Transformer	95.59	Custom clinical	EEG spectrogram

al. reporting the best performance of 95.59% [34]. Boosting-based approaches also show strong performance, with an accuracy of 90.80%. Models such as XGBoost techniques consistently achieve accuracy values above 90% [35]. Deep learning models demonstrate stable performance (mean: 93.5%, range: 88.0–98.86%) [37] [24], indicating that neural architectures are effective in learning spatial and temporal patterns from EEG signals. Traditional machine learning approaches report comparatively lower performance, with Guo et al. [36] reporting 89.1% using an autoencoder-based SVM model.

The higher performance of advanced architectures compared to traditional machine learning highlights the strength of modern neural models in capturing complex EEG patterns related to ADHD. Boosting methods offer a good balance between accuracy and interpretability, making them suitable for clinical use. Deep learning approaches also show strong performance by automatically learning meaningful representations from EEG data, reducing reliance on manual feature engineering.

C. Interpretation and Implications for ADHD Biomarker Development

The ranking of machine learning approaches (Advanced > Deep learning> Boosting> Traditional) provides useful insight for selecting appropriate methods for EEG-based ADHD biomarker development:

1) **For Clinical Translation:** For clinical use, a combined strategy is beneficial. Boosting methods such as XGBoost [35] provide interpretable feature importance, helping clinicians understand key EEG markers despite moderate accuracy (90.80%) [22] [23]. Deep learning models such as CNN–LSTM [30] [37] achieve higher accuracy but are less interpretable. Advanced architectures, including transformer-based models [34], show strong performance but also require improved explainability. Therefore, using boosting for transparent screening and deep learning for final classification offers a practical approach for clinically applicable ADHD diagnosis [32] [39].
2) **For Research Advancement:** Advanced architectures, particularly Alsharif et al. [34] and transfer learning–based approaches [38], demonstrate the highest performance and enable modelling of complex spatial and temporal EEG patterns. These approaches are valuable for gaining deeper insights into the neurophysiological mechanisms associated with ADHD.
3) **For Resource-Constrained Settings:** Traditional machine learning models, despite lower average accuracy (89.1%), remain useful in scenarios where computational efficiency and simplicity are important. Such models may be suitable for real-time applications, portable EEG devices, or preliminary screening systems.

VI. FUTURE DIRECTION

Advancement in large-scale datasets, computing resources, and EEG technology are creating new opportunities for ADHD research. Future progress depends on improving model generalizability, integrating multimodal data, and strengthening methodological rigor. Priorities include combining EEG datasets from various sources, conducting longitudinal studies, and combining EEG with complementary modalities such as MRI, genetics, eye tracking, and behavioral assessments to better understand ADHD mechanisms. Wearable EEG systems enable more naturalistic monitoring, while interpretable machine learning methods improve clinical trust and usability [39]. Enhancing reproducibility also requires standardized preprocessing practices, transparent reporting, and open sharing of data and code using established tools such as EEGLAB [28] and the PREP pipeline [29]. These directions can help translate EEG research into clinically useful tools for ADHD diagnosis and personalized treatment.

VII. CONCLUSION

EEG research has provided valuable insights into the neural mechanisms of ADHD, highlighting alterations in spectral activity, event-related responses, and large-scale brain connectivity. These findings indicate that difficulties in attention regulation, executive functioning, and inter-regional communication are key features of the disorder. However, EEG-based biomarkers have not yet achieved sufficient reliability for routine clinical diagnosis, mainly due to variations in study design, participant characteristics, and analysis methods.

Future progress will require larger collaborative datasets, improved standardization of experimental and preprocessing procedures, and better integration of EEG with complementary data sources. Although recent advances in deep learning and transformer-based models show promising results, further validation across diverse populations is needed before clinical adoption. Overall, EEG remains an important tool for understanding ADHD and holds strong potential for future use in continuous monitoring and personalized treatment strategies.

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